Replication and Extension

Partisan Conflict over Content Moderation Is More Than Disagreement about Facts

Appel, Pan & Roberts (2023), Science Advances

Sarah Novotny | POP7004 Applied Statistics II | Trinity College Dublin

FINAL EDITION

The Times

EXTRA! EXTRA!

CANDIDATES IN DEAD HEAT

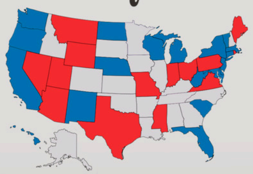
Control of Congress Hangs in Balance

The control of power in both the United States House of Representatives and Senate hung in the balance this morning as election returns from across the nation poured in.

With an unexpectedly strong showing from challengers in traditional party strongholds, the partisan map will have to be redrawn. It is highly likely, however, that the next Senate will be evenly split. This would represent the first such tie in the U.S. Senate since the late 1950's, making the possibility of Congressional gridlock all but an inevitable reality for the upcoming term.

Additionally, the races were also incredibly tight across the nation,

State by State



Governor Wins Second Term

For only the second time in the last twenty-four years, a governor has won re-election in what is generally regarded as the toughest state

Incumbents Lead Local Elections

Local voters overwhelmingly cast their ballots for the incumbents, yesterday. All three local members of the U.S. House of Representatives retained their respective seats.

Additionally, in keeping with recent trends, all local members of the state legislature will be going back to the state capital for an another term.

The only elected official who won't be returning next January will be one county Supervisor who has been embroiled in an ethics scandle since soon after his re-election four years ago.

The Mayor's race was all but decided months ago when no significant challenger surfaced, thus paving the way for an easy re-election. Despite having no real opposition, the Mayor stressed that he was

The Original Experiment

Survey experiment:

1,120 U.S. respondents

(673 Dem, 447 Rep)

2 political headlines:

(1 pro-Dem, 1 pro-Rep)

Key design:

headlines crossed with party

aligned vs. misaligned conditions

Method:

Linear Probability Model (WLS), survey weights, clustered SEs

Each respondent evaluates 2 headlines × 3 outcomes = 6 responses

Outcome	Pro-Democrat Headline	Pro-Republican Headline
Remove	Should it be taken down?	Should it be taken down?
Harm	Is it harmful?	Is it harmful?
Censorship	Would removal be censorship?	Would removal be censorship?

Investigations

party promotion - a desire to leave misinformation online that promotes one's own party

preference gap - differences in internalized preferences about whether misinformation should be removed.

To challenge the current explanations

fact gap - differences in perceptions about what is misinformation.

Core Findings

Large partisan baseline gaps:

- Democrats ~35pp more likely to favor removal than Republicans
- Republicans ~36pp more likely to call removal censorship

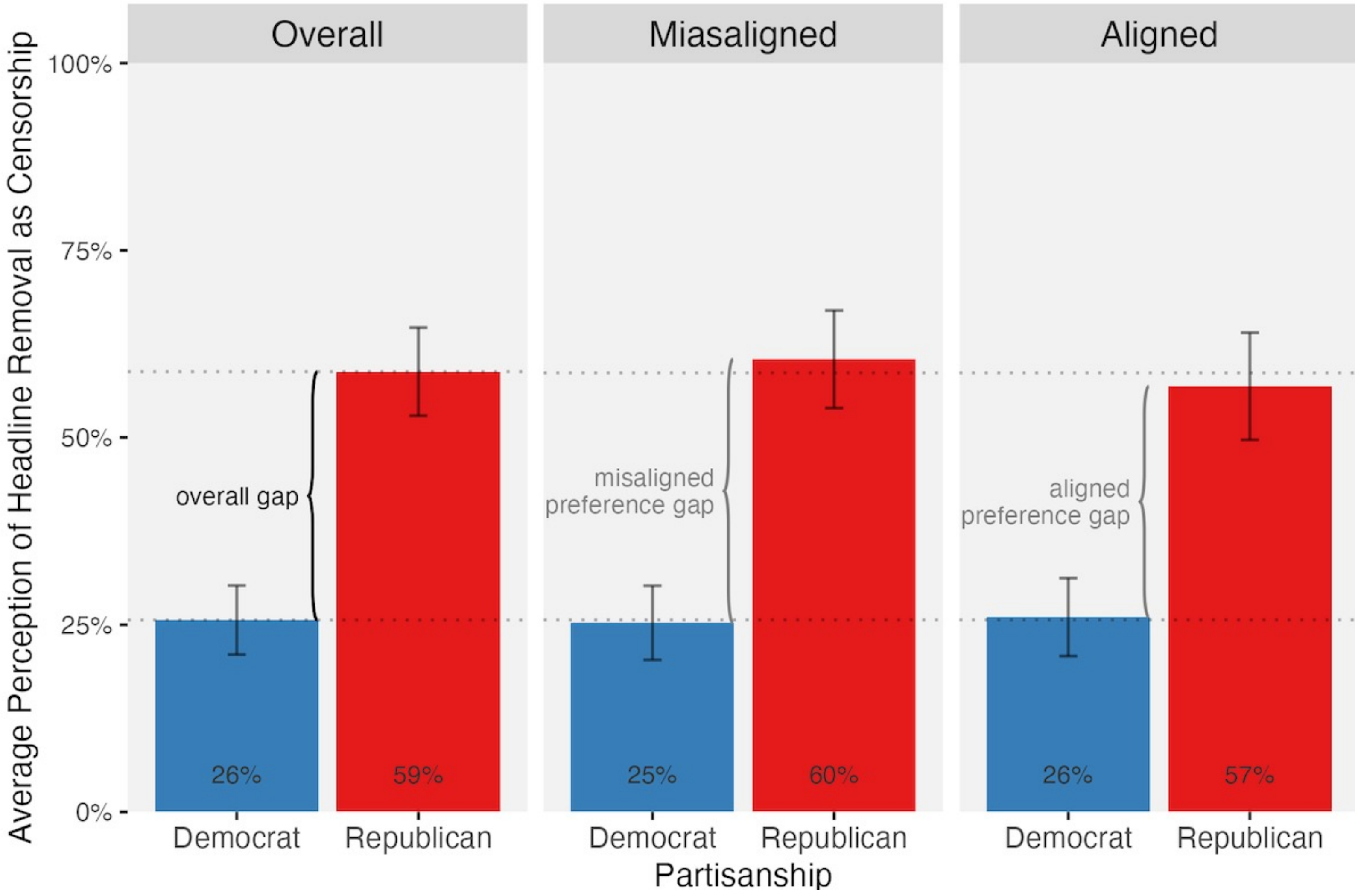
Party promotion: Democrats less willing to remove co-partisan headlines

- Dem x Pro-Dem interaction: -11pp ($p < 0.001$) for removal
- Republican promotion: small and statistically insignificant

Critical finding: gaps persist even when both parties agree a headline is false

Partisan conflict over moderation is not reducible to factual disagreement

Gaps persist even when both parties agree a headline is false



The Twist: LPM vs. Logistic Regression

The paper uses the Linear Probability Model (OLS for binary outcomes)

- Pro: coefficients directly in percentage-point units
- Con: predicted probabilities can fall outside [0, 1]

Extension: re-estimate with logistic regression (GLM, binomial, logit link)

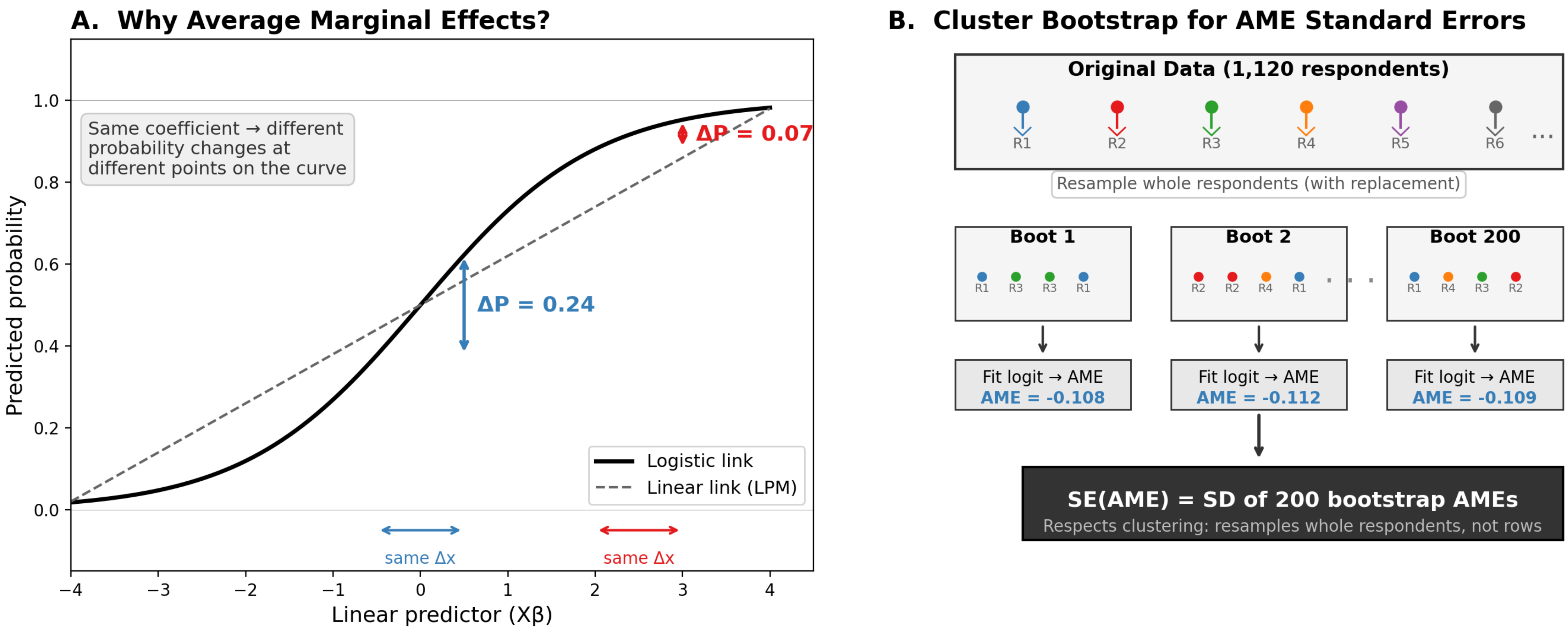
Compare LPM coefficients to logit average marginal effects (AMEs)

- AME: finite-difference approach, standard errors via cluster bootstrap

Also compared: predicted probabilities, boundary violations, odds ratios

Question: do the substantive conclusions depend on the estimation method?

Average Marginal Effects & Cluster Bootstrap



$$\mathcal{L} = \oint E_A t$$

$$\nabla \cdot E = 0 \qquad \nabla \times E = -\frac{1}{c}\frac{\partial H}{\partial t}$$

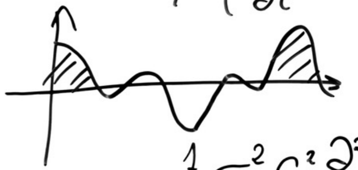
$$\nabla \cdot H = 0 \qquad \nabla \times H = \frac{1}{c}\frac{\partial E}{\partial t}$$

$$f(\omega) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \omega}\, dx \quad \frac{dt}{d\theta}$$

$$i\hbar \frac{\partial}{\partial t}\Psi = H\Psi$$

$$\rho\left(\frac{\partial v}{\partial t} + v \cdot \nabla v\right) = -\nabla p + \nabla \cdot T + f$$

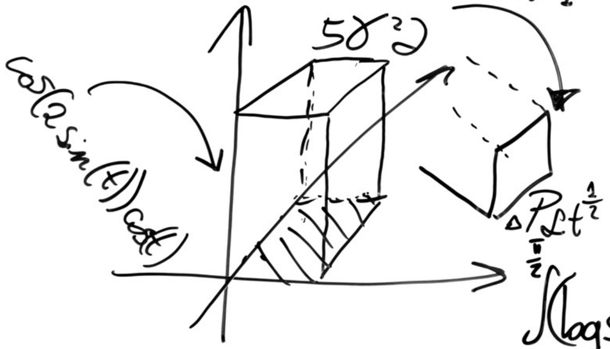
$$+ \sum_{i=1}^{n} \frac{q_i}{2} H_i^M + C_s \frac{D}{Q} + C_0 D + \frac{Q(p-D)}{2p} H^M + F_0 N + F_0 N + \sum_{i=1}^{n} D_i w_i d_i \frac{(1+\alpha)}{F_x^{FTL}}$$



$$\boxed{H = -\sum p(x) \log p(x)}$$

$$\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} + \frac{\partial V}{\partial t} - rV = 0$$

$$TC(Q, q_i, m_i) = \sum_{i=1}^{n}\left[\frac{D_i}{m_i q_i} S_i + c_i^v D_i + \frac{q_i H_i^v}{2}\left(m_i\left(1 - \frac{D_i}{P_i}\right) - 1 + 2\frac{D_i}{P_i}\right)\right] +$$



$$\begin{bmatrix} \dfrac{d\Delta p(s,\phi)}{d\phi} \\ \dfrac{d\Delta M(s,\phi)}{d\phi} \end{bmatrix} = \begin{bmatrix} \gamma & -\mathcal{L} \\ -\beta & 0 \end{bmatrix} \begin{bmatrix} \Delta p(s,\phi) \\ \Delta M(s,\phi) \end{bmatrix}$$

$$\int_0^{\frac{\pi}{2}} (\log \sin x)^2\, dx = \int_0^{\frac{\pi}{2}} (\log \cos x)^2\, dx = \frac{\pi}{2}\left\{\frac{\pi^2}{12} + (\log 2)^2\right\}$$

LPM vs. Logit: Party Promotion Effects

Average marginal effects on the probability scale (base specification)

Outcome	Term	LPM Coef	LPM SE	Logit AME	AME SE	Diff
Remove	Dem x Pro-Dem	-0.110	0.023	-0.110	0.024	0.000
Remove	Rep x Pro-Rep	0.002	0.023	0.002	0.023	0.000
Harm	Dem x Pro-Dem	-0.130	0.027	-0.130	0.028	0.000
Harm	Rep x Pro-Rep	0.033	0.027	0.033	0.026	0.000
Censorship	Dem x Pro-Dem	0.013	0.020	0.013	0.020	0.000
Censorship	Rep x Pro-Rep	-0.002	0.027	-0.002	0.025	0.000

LPM vs. Logit: Party Promotion Effects

Average marginal effects on the probability scale (base specification)

Outcome	Term	LPM Coef	LPM SE	Logit AME	AME SE	Diff
Remove	Dem x Pro-Dem	-0.110	0.023	-0.110	0.024	0.000
Remove	Rep x Pro-Rep	0.002	0.023	0.002	0.023	0.000
Harm	Dem x Pro-Dem	-0.130	0.027	-0.130	0.028	[illegible]
Harm	Rep x Pro-Rep	0.033	0.027	0.033	0.026	[illegible]
Censorship	Dem x Pro-Dem	0.013	0.020	0.013	0.020	0.000
Censorship	Rep x Pro-Rep	-0.002	0.027	-0.002	0.025	0.000

EXACTLY THE SAME 🤔

Note: Base model is saturated (4 parameters, 4 cells), so LPM and logit AMEs match exactly.

Why the Base Model Shows Zero Difference

Saturated vs. non-saturated models and the role of the link function

The base model has 4 parameters for 4 cells (2 parties × 2 alignments)

Any saturated model reproduces exact cell means regardless of link function

LPM line and logit curve both pass through the same 4 points — no room to diverge

Adding demographic controls (age, gender, education, income, race, …)

- Breaks saturation: 14 parameters, still only 4 party–alignment cells
- The link function must now interpolate across covariate values
- Linear vs. logistic links can produce different predictions

Result: with controls, predictions diverge — but only by < 0.6 percentage points

Outcomes in the 25–75% range sit where logistic and linear links are nearly parallel

Predicted Probabilities (with Controls)

Non-saturated specification — link function matters

Outcome	Scenario	LPM	Logit	Diff (pp)
Remove	Dem, Misaligned	0.761	0.763	-0.25
Remove	Dem, Aligned	0.641	0.643	-0.18
Remove	Rep, Misaligned	0.340	0.337	+0.36
Remove	Rep, Aligned	0.349	0.345	+0.33
Harm	Rep, Misaligned	0.232	0.226	+0.59
Censorship	Dem, Misaligned	0.245	0.241	+0.41

Maximum divergence: 0.59pp. Zero LPM boundary violations.
LPM and logit yield substantively identical conclusions for this application.

Logit Odds Ratios: A Complementary Lens

Democrat party promotion (Remove):

- OR = 0.60 — when a headline favors their own party, Democrats have 40% lower odds of wanting it removed
- They protect co-partisan content: 75% want cross-partisan headlines removed, but only 64% for co-partisan

Republican party promotion (Remove):

- OR = 1.01 — Republicans show virtually no difference based on headline alignment
- Consistent with the near-zero LPM coefficient (0.002, $p = 0.93$)

Partisan baseline gap (Remove):

- OR = 0.18 — Republicans have 82% lower odds than Democrats of favoring removal overall

Odds ratios restate the LPM findings multiplicatively — no change in conclusions

Conclusions

1. Replication confirms all core findings of Appel, Pan & Roberts (2023)

   Republicans are half as likely as Democrats to say that the content should be removed
   & >2x as likely to consider removal as censorship – even when they agree with the content.

2. LPM vs. logit comparison:

   The paper's choice of LPM is well-justified for these outcomes

   < 1pp divergence even with demographic controls

   Zero boundary violations in the LPM

3. When would logit matter more?

   Outcomes with base rates near 0% or 100%

   Models with many covariates pushing predictions to extremes

   Neither condition applies here (base rates 25–75%)

Replication Challenges

1. ggbrace package incompatibility

    Original used geom_brace() from ggbrace v0.1.0 (2022)

    Current versions removed this function; ggplot2 4.x broke the old version via S7

    Solution: compatibility shim translating old API to stat_brace()/stat_bracetext()

2. Missing output directories

    figures/paper/ subdirectory not documented in README

3. Data type handling in the twist

    Original vec.controls includes race as character — needed mode() not mean()

    Survey weights interact differently with GLM cluster SEs (statsmodels warns too)